

"Review of thermodynamics"

Definitions

E : Internal energy (J)

H : Enthalpy (J)

S : Entropy ($\frac{J}{K}$)

m : mass of gas (kg)

n : moles (kmol)

$\bar{M}$: molar mass ($\frac{kg}{kmol}$)

$$\bar{M} = \frac{m}{n} \left(\frac{kg}{kmol} \right)$$

e : mass-specific internal energy ($\frac{J}{kg}$)

h : " " Enthalpy ($\frac{J}{kg}$)

s : " " Entropy ($\frac{J}{kgK}$)

$$e = \frac{E}{m}$$

$$h = \frac{H}{m}$$

$$s = \frac{S}{m}$$

→ mass-specific

$$\bar{e} = \frac{E}{n} \left(\frac{J}{kmol} \right)$$

$$\bar{h} = \frac{H}{n} \left(\frac{J}{kmol} \right)$$

$$\bar{s} = \frac{S}{n} \left(\frac{J}{K kmol} \right)$$

→ mol-specific

connection between mol-specific & mass specific

$$\bar{e} = \bar{M} e$$

$$\bar{h} = \bar{M} h$$

$$\bar{s} = \bar{M} s$$

constant volume specific heat ($c_v, \bar{c}_v$)

" pressure " " ($c_p, \bar{c}_p$)

$$c_v = \left(\frac{\partial e}{\partial T} \right)_v \left(\frac{J}{kgK} \right), \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p \left(\frac{J}{kgK} \right)$$

(P-2)

$$\bar{c}_v = \bar{M} c_v = \left(\frac{\partial \bar{e}}{\partial T} \right)_v \left(\frac{J}{\text{kmol} \cdot K} \right), \quad \bar{c}_p = \bar{M} c_p = \left(\frac{\partial \bar{h}}{\partial T} \right)_p \left(\frac{J}{\text{kmol} \cdot K} \right)$$

Specific heat ratio: $\gamma = \frac{\bar{c}_p}{\bar{c}_v} = \frac{\bar{M} c_p}{\bar{M} c_v} = \frac{c_p}{c_v}$

Perfect (ideal) gas

$Pv = n \bar{R} T \rightarrow$ Eqⁿ of state, $\bar{R}$: universal gas constant

$$\bar{R} = 8314.3 \frac{J}{\text{kmol} \cdot K} = 8.3143 \frac{J}{\text{mol} \cdot K}$$

$$\frac{Pv}{n} = \frac{n \bar{R} T}{n} \Rightarrow \frac{P}{\frac{1}{v}} = \frac{\bar{R}}{\bar{M}} T = R T \quad (\bar{R} = \bar{M} R) \Rightarrow P = \gamma R T$$

R : specific gas constant.

$$\bar{M}_{H_2} = 2 \frac{\text{kg}}{\text{kmol}} \Rightarrow R_{H_2} = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{2} \frac{J/\text{kmol} \cdot K}{\text{kg}/\text{kmol}} = 4157.2 \frac{J}{\text{kg} \cdot K}$$

$$\bar{M}_{N_2} = 28 \frac{\text{kg}}{\text{kmol}} \rightarrow R_{N_2} = \frac{\bar{R}}{\bar{M}} = 296.9 \frac{J}{\text{kg} \cdot K}$$

$$\bar{M}_{CO_2} = 44 \frac{\text{kg}}{\text{kmol}} \rightarrow R_{CO_2} = 189 \frac{J}{\text{kg} \cdot K}$$

$$\bar{M}_{H_2O} = 18 \frac{\text{kg}}{\text{kmol}} \rightarrow R_{H_2O} = 461.9 \frac{J}{\text{kg} \cdot K}$$

For perfect gas, $e = e(T)$ & $h = e + \frac{P}{\gamma} = e + RT = h(T)$

③

$\rightarrow$ depend only on (T)

$$c_v = \left(\frac{\partial e}{\partial T} \right)_v = \frac{de}{dT} \quad \gamma = \frac{c_p}{c_v}$$

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p = \frac{dh}{dT}$$

$$h = e + R T$$

$$\frac{dh}{dT} = \frac{de}{dT} + R$$

$$c_p = c_v + R$$

$$\frac{c_p}{c_v} = 1 + \frac{R}{c_v} \Rightarrow$$

$$\begin{cases} c_p = \frac{\gamma R}{\gamma - 1} \\ c_v = \frac{R}{\gamma - 1} \end{cases}$$

$$\text{Similarly: } \bar{c}_v = \frac{d\bar{e}}{dT}, \quad \bar{c}_p = \frac{d\bar{h}}{dT}, \quad \gamma = \frac{\bar{c}_p}{\bar{c}_v}, \quad \bar{c}_p = \frac{\gamma}{\gamma - 1} \bar{R}$$

$$\bar{c}_v = \frac{\bar{R}}{\gamma - 1}, \quad \bar{c}_p = \bar{c}_v + \bar{R}$$